hw 3, env s 193ds

Damian Rojas

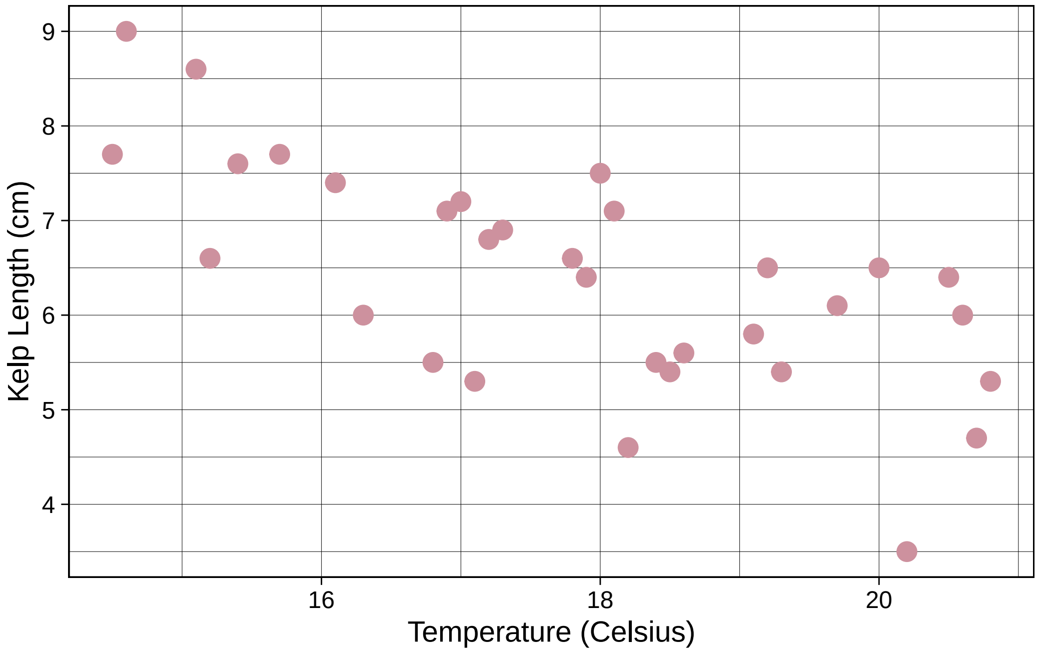
```
#read packages
library(tidyverse)
library(janitor)
library(here)


#store kelp data
kelp <- read.csv("data/temp-kelp.csv")

#store personal data
walks <- read.csv("data/walks_data.csv")
```

1)

a)

The appropriate tests we can run to test the strength of the relationship between temperature and giant kelp frond elongation rate are a Pearson's correlation test and a Spearman's rank correlation. The differences between the two tests are that a Pearson's test observes the strength between linear relationships while the Spearman's test observes the strength between monotonic relationships that do not need to be linear.

b) Visualizations

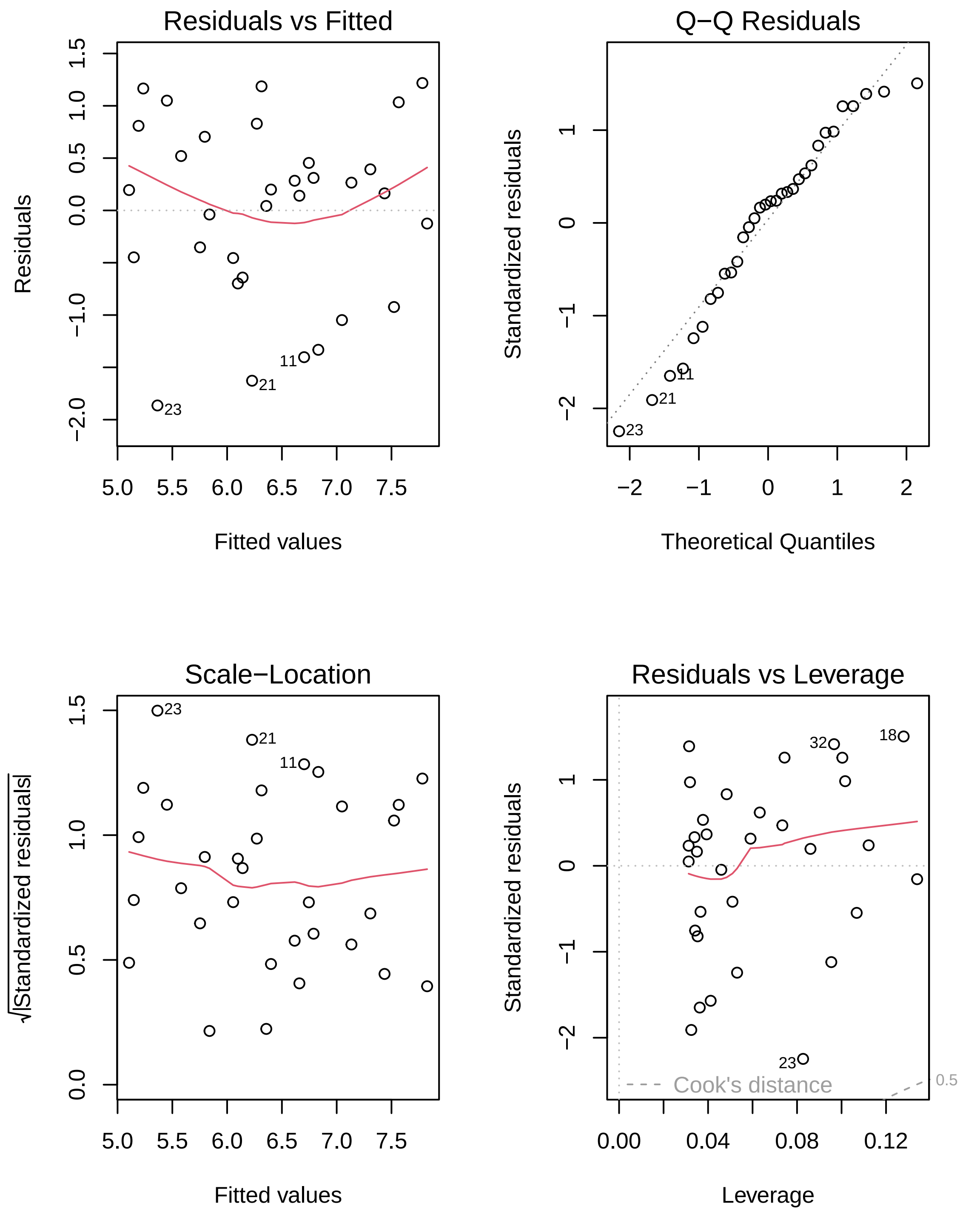


c) Checking Assumptions / Run test

```
#create linear model for kelp data
kelp_model <- lm (kelp_elong ~ temp_c, data = kelp)

#show lm for kelp
kelp_model
```

```
Call:
lm(formula = kelp_elong ~ temp_c, data = kelp)

Coefficients:
(Intercept)       temp_c
    14.0864      -0.4318
```

```
#kelp elongation as function of temp
#
#plotted as 2x2 grid
 par(mfrow = c(2,2))
 #summary plots of kelp model
plot (kelp_model)
```


Residuals vs Fitted
Residuals
Fitted values
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
0.5

Checked normality, homoscedasticity, and any outliers of data with summary plots. Found to be normal, linear, with no outliers affecting data.

```
#pearsons cor test is the method for linear models
#run test with x as kelp elongation, y as temp, method as pearson
cor.test(x = kelp$temp_c, y = kelp$kelp_elong, method = "pearson")
```

```

	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor 
-0.6877489 
```

d)

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a Pearson's correlation test. We found a strong negative relationship between kelp elongation and temperature (Pearson's r = -0.69, t(30) = -5.19, $p < 0.001$), where temperature significantly predicted kelp elongation length ($R^2 = 0.476$). We reject the null hypothesis due to the p-value being less than $\alpha = 0.05$.

e)

The Pearson's correlation shows a statistically significant negative relationship between kelp elongation and temperature. This shows that an increase in ocean or seawater temperature causes kelp growth to stunt, leading to shorter kelp within these affected regions. Further implications of this phenomena may lead to giant kelp height remaining at these shortened heights unless resistant regions are studied and found to have any possible mediation for non-resistant groups.

f)

```r
#run spearmans rank test
cor.test( x = kelp$temp_c, y = kelp$kelp_elong, method = "spearman")
```

```
Warning in cor.test.default(x = kelp$temp_c, y = kelp$kelp_elong, method =
"spearman"): Cannot compute exact p-value with ties
```

```

	Spearman's rank correlation rho

data:  kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.6891684
```

The Spearman's test shows the same p-val as the Pearson's test leading me to the same conclusion (rejecting the null hypothesis). The change in test type keeps me from claiming a linear relationship due to the restrictions behind a Spearman's test.

2a) Personal Data

```r
#from HW 2
walks |>
mutate(work_due = as_factor(work_due))
```

```
   DATE WEATHER_F TIME.Start. WIND.SPEED WIND.DIR. DEADTIME.min.sec. TIME.End.
1  4/28        66     7:06 PM          4      AWAY              0:00   7:22 PM
2  4/29        64     5:54 PM          4   AGAINST              0:46   6:09 PM
3  4/30        62     7:29 PM          8      AWAY              0:49   7:44 PM
4   5/3        60     7:23 PM         11   AGAINST              0:14   7:38 PM
5   5/4        60        7:07          8   AGAINST              0:00      7:28
6   5/6        64        8:04          6      AWAY              0:16      8:18
7  5/12        60        7:42          2   AGAINST              0:24      7:57
8  5/12        60        8:05          2      AWAY              0:00      8:22
9  5/13        65        6:37          6   AGAINST              0:15      6:52
10 5/13        63        7:16          5      AWAY              0:07      7:31
11 5/14        66        3:43          6      AWAY              0:00      3:55
```

```
12 5/15        64       1:58         7    AGAINST              0:00       2:12
13 5/19        73       6:44         8    AGAINST              0:27       6:59
14 5/19        72       7:11         6       AWAY              0:19       7:25
15 5/20        67       8:07         4    AGAINST              1:00       8:22
16 5/20        67       8:26         4       AWAY              0:00       8:40
   work_due walk_time walk_time_s
1         1     15:48         948
2         0     14:10         850
3         2     14:38         878
4         1     14:43         883
5         0     14:28         868
6         1     13:42         822
7         1     14:37         877
8         1     16:14         974
9         0     15:13         913
10        0     15:12         912
11        2     12:54         774
12        5     13:59         839
13        0     14:32         872
14        0     14:11         851
15        0     14:07         847
16        0     14:42         882
```

```r
       #factor so data is listed in order of appearance, cleans up repeats
#create graph, start with gg plot as base layer
# x = categorical predictor
ggplot(walks, aes(x = work_due,
# y = response variable (in seconds to avoid graph looking skewed)
                  y = walk_time_s)) +
# scatter plot to compare relationship between predictor and response variable
geom_jitter(height = 0,
            width = 0,
            shape = 16,
            size=5,
#change color
           color="purple4") +

#label graph
labs(title="Workload vs Walk time",
     x = "Assignments due(#)",
     y = "Length of walk (sec)") +
#change theme
```

```r
theme_linedraw()
```

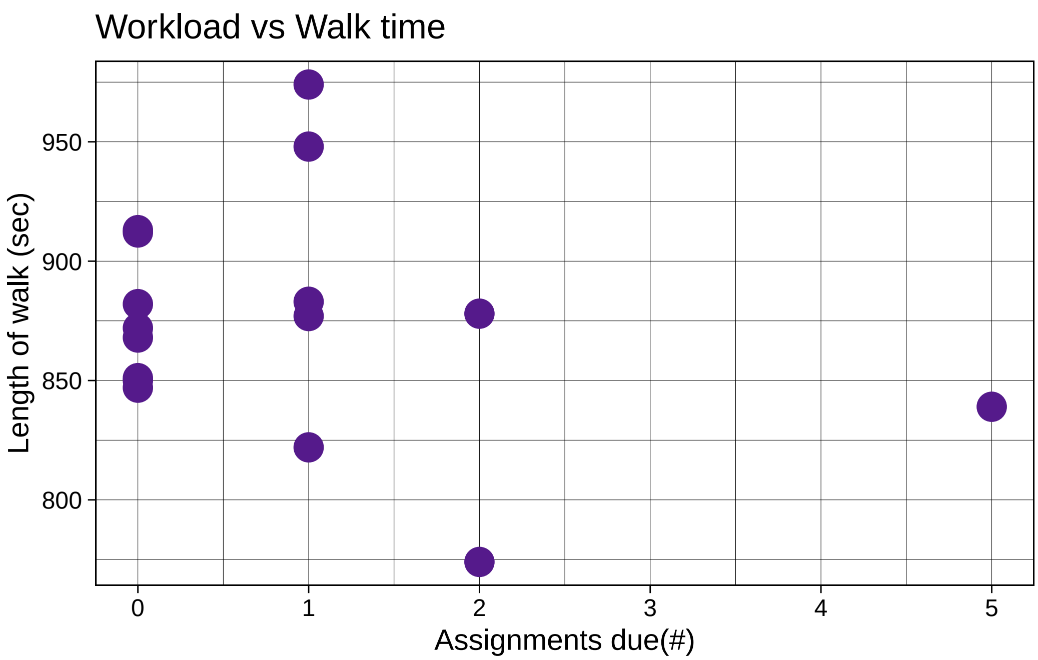


Fig 1

Scatterplot showing relationship between number of assignments due and length of walk in seconds. Each data point is represented by a purple point (n = 16).

```r
#from HW2 to show potential relationship between degrees and walk time
#collect personal data
walks |>

mutate(WEATHER_F = as_factor(WEATHER_F))
```

	DATE	WEATHER_F	TIME.Start.	WIND.SPEED	WIND.DIR.	DEADTIME.min.sec.	TIME.End.
1	4/28	66	7:06 PM	4	AWAY	0:00	7:22 PM
2	4/29	64	5:54 PM	4	AGAINST	0:46	6:09 PM
3	4/30	62	7:29 PM	8	AWAY	0:49	7:44 PM
4	5/3	60	7:23 PM	11	AGAINST	0:14	7:38 PM
5	5/4	60	7:07	8	AGAINST	0:00	7:28
6	5/6	64	8:04	6	AWAY	0:16	8:18
7	5/12	60	7:42	2	AGAINST	0:24	7:57

```
8  5/12        60        8:05          2      AWAY                0:00       8:22
9  5/13        65        6:37          6    AGAINST               0:15       6:52
10 5/13        63        7:16          5      AWAY                0:07       7:31
11 5/14        66        3:43          6      AWAY                0:00       3:55
12 5/15        64        1:58          7    AGAINST               0:00       2:12
13 5/19        73        6:44          8    AGAINST               0:27       6:59
14 5/19        72        7:11          6      AWAY                0:19       7:25
15 5/20        67        8:07          4    AGAINST               1:00       8:22
16 5/20        67        8:26          4      AWAY                0:00       8:40
   work_due walk_time walk_time_s
1         1     15:48         948
2         0     14:10         850
3         2     14:38         878
4         1     14:43         883
5         0     14:28         868
6         1     13:42         822
7         1     14:37         877
8         1     16:14         974
9         0     15:13         913
10        0     15:12         912
11        2     12:54         774
12        5     13:59         839
13        0     14:32         872
14        0     14:11         851
15        0     14:07         847
16        0     14:42         882
```

```r
#create graph, start with gg plot as base layer
#x=continuous predictor, weather
#y= length of walk
#take degrees F ofdatqa sheet, abel in axis
ggplot(walks, aes(x = WEATHER_F,
                  y = walk_time_s)) +
#scatter plot to compare relationship between continuous predictor and response
geom_point(color="green3", size = 5) +

#label graph
labs(title = "Weather vs Walk Time",
     x = "Weather(°F)",
     y = "Length of Walk (sec)") +

#change theme
```

```r
theme_bw()
```

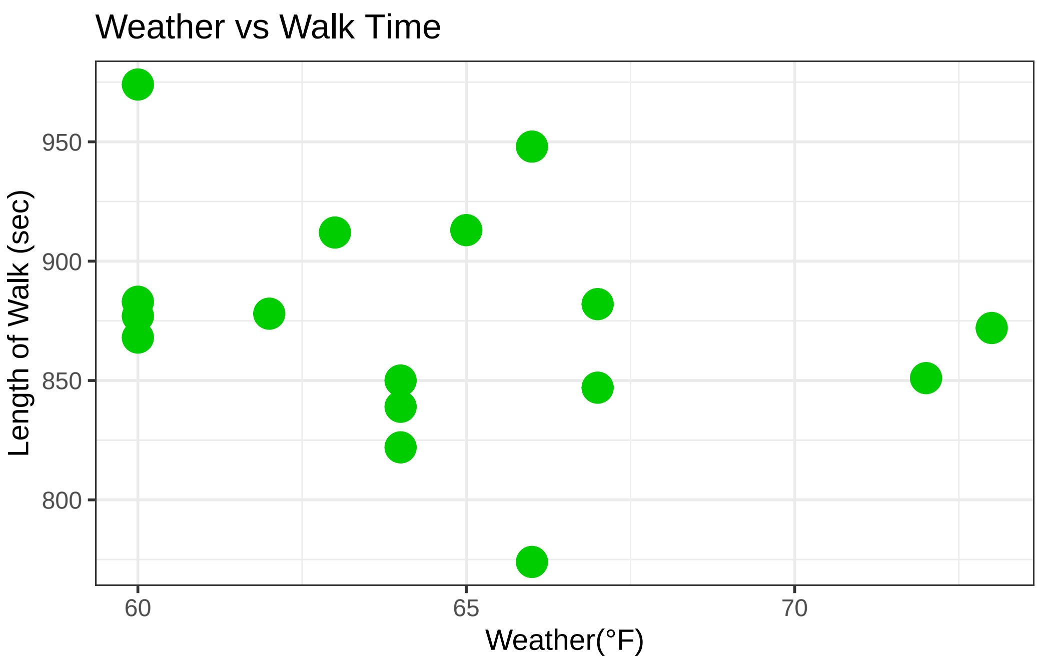


Fig 2

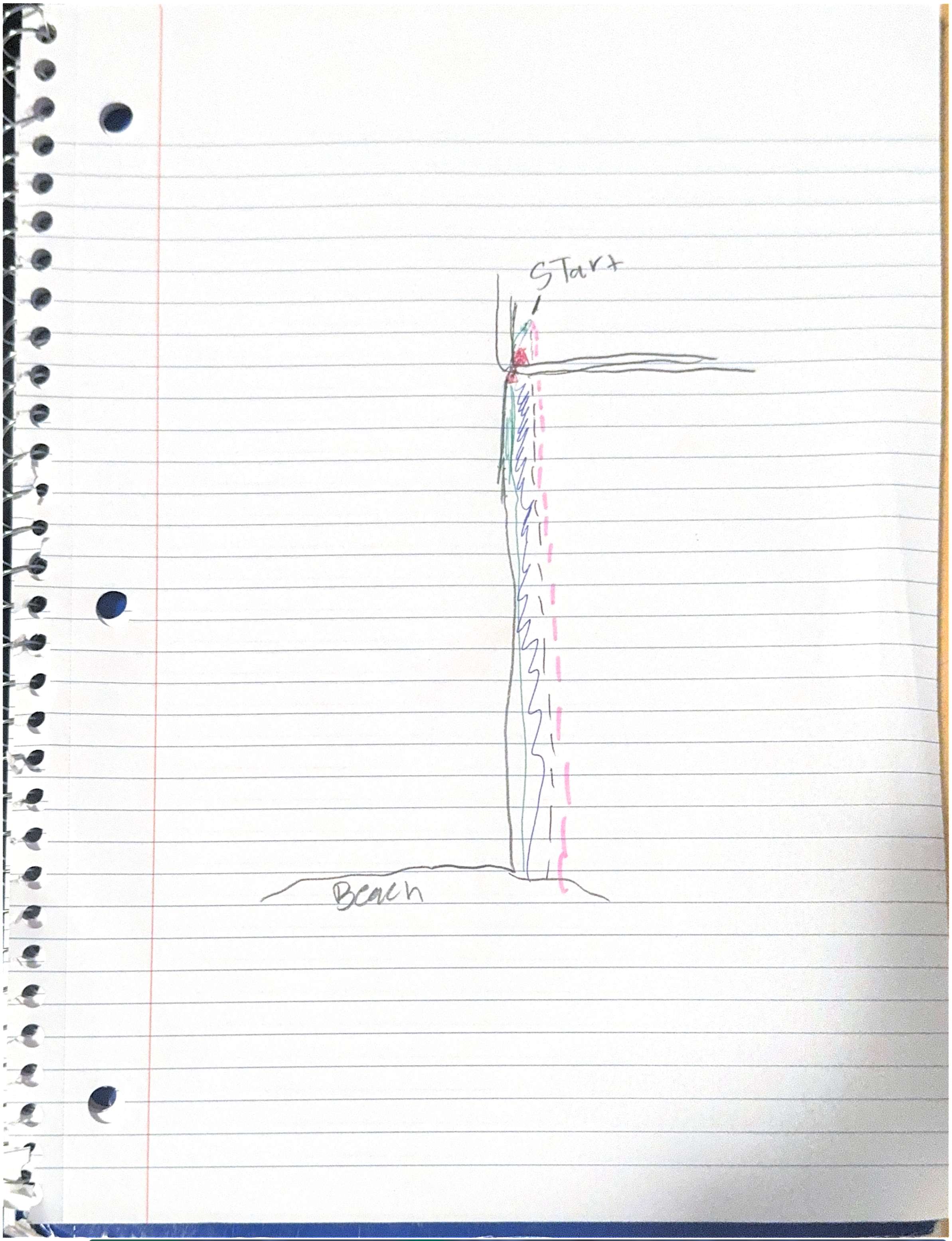
Scatterplot showing relationship between weather (in degrees Fahrenheit) and length of walk (in seconds) with each green point representing an individual walk (n=16).

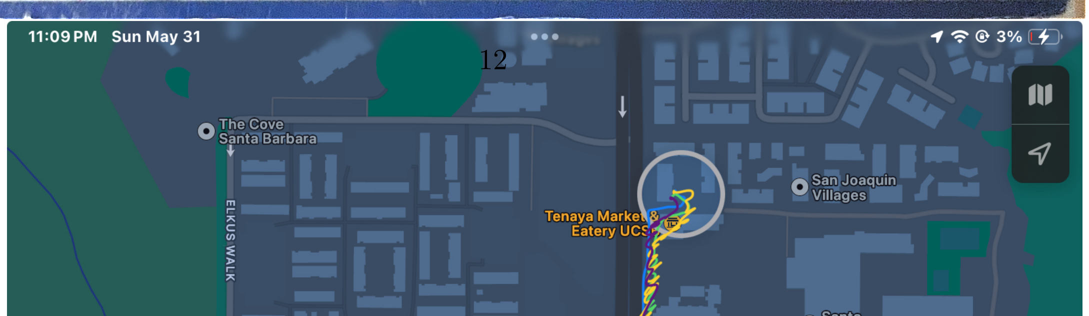
3

a)

An affective visualization for this project could look like the route of the walk I would take during the personal data project. I could differentiate the route tracking by altering line color/width depending on assignment due. I could also include red stop signs or another visualization to illustrate where the walks would typically get slowed down.

b)


Start
Beach


11:09 PM Sun May 31
3%
The Cove
Santa Barbara
ELKUS WALK
San Joaquin
Villages
Tenaya Marke
Eatery UCS

d)

This piece shows the route of the walk I would take along with a display of the various different workloads I had while taking the walk. I was influenced by Jill Pelto's artwork displaying scenic backgrounds which provided information to the data she was displaying. I took inspiration by displaying (albeit not art) the route to help show the visualization of work due and walk time. The process for my draft was simply taking a screenshot of the route (for simplicty) and adding the various lines (with different colors and shape) to help differentiate between the various workloads.

e

Link to slideshow for workshop 9

4

a)

The test the authors selected was a t-test, more specficially for this article the test is an unpaired Welch's two-sample t test. The response variable for this experiment is the AQI (air quality index) of 4 sites encompassing major cities in India (ITO-Delhi, Worli-Mumbai, Jadavpur-Kolkata, and Manali Villiage-Chennai) while the predictor variable is COVID-19 lockdown confinement (Reduced human activity).

Table 2 Welch's two-sample *t* test analysis

From: Air quality assessment among populous sites of major metropolitan cities in India during COVID-19 pandemic confinement

	Site 1		Site 2		Site 3		Site 4	
	Sample A	**Sample B**	**Sample A**	**Sample B**	**Sample A**	**Sample B**	**Sample A**	**Sample B**
Mean	241.65	134.25	159.12	65.77	144.86	57.45	75.78	63.20
Observations	36	35	36	35	36	35	36	35
Hypothesized mean difference	0		0		0		0	
Degree of freedom	62		38		40		42	
95% confidence interval	(71.05, 143.75)		(69.45, 116.99)		(63.14, 111.65)		(1.09, 24.05)	
t-statistic	5.91		7.94		7.28		2.20	
P ($T \leq t$) one-tail	0.00000015		0.0000000014		0.0000000074		0.03	
t Critical one-tail	1.67		1.68		1.68		1.68	

b) The table shows the data in a clear way, however if you are just reading the table without looking at the article it is hard to comprehend. The table does not share what the values mean in regards to the article, but they just show what the values are.

c)

The authors handled visual clutter very well for the table. There is bolded text reference table number and table title, along with what the values of the table represent. Everything is spaced out very well making the data very easy to read.

d)

As referenced in part B, the table data is not very clear.The mean AQI is listed as mean, however if you had not read the article you would not be able to tell what the mean is referencing. WHen looking back on the table for this assignment I asked myself “mean what?” which shows how the table does not stand on its own.